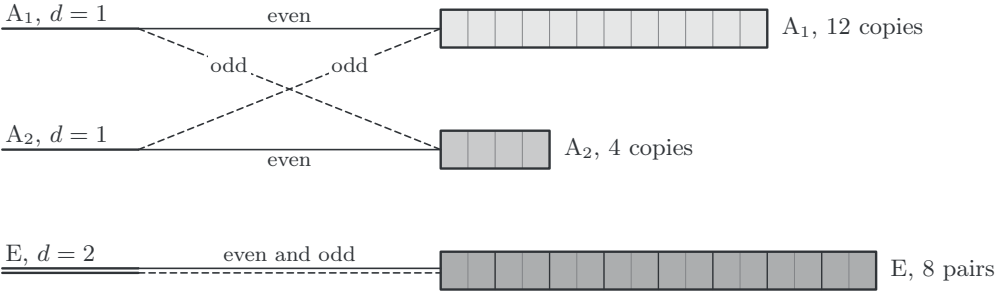


$\mathcal{H}_{\text{spin}} = (\mathbb{C}^2)^{\otimes 5}$ for the five protons, $\dim = 32$; under G_6 it holds 12 copies of A_1 , 4 of A_2 and 8 of E, so $12 + 4 + 2 \cdot 8 = 32$; at right each spin species is a bar of one unit per dimension

spatial species Γ , dimension d_Γ

spin species, copies in $\mathcal{H}_{\text{spin}}$



the weight of a level is the length of the bar it is joined to: $w_{A_1}(\pm) = 12, 4$; $w_{A_2}(\pm) = 4, 12$; $w_E = 8$ per member

solid: even parity, $\chi_+ = A_1$; dashed: odd parity, $\chi_- = A_2$; a level pairs with the spin species whose product with it contains $\chi_\pm$